[INCH-POUND]
A-A-3085
November 7, 1996

COMMERCIAL ITEM DESCRIPTION

PROTRACTORS, MECHANIC'S, UNIVERSAL BEVEL AND VERNIER SCALE

The General Services Administration has authorized the use of this Commercial Item Description for the procurement of the type I, class 1, style B protractor specified in GGG-P-676 for all federal agencies.

1. SCOPE.

1.1 Scope. This Commercial Item Description covers protractors designed for precision measuring, checking, laying out of angular work, and scribing layout lines.

2. CLASSIFICATION.

2.1 Classification. Not applicable.

3. SALIENT CHARACTERISTICS.

3.1 Design. The universal bevel, vernier scale protractor shall have a frame, graduated dial, thumbnut, and adjustable blade with an acute-angle attachment and revolving dial, and shall be similar to Figure 1. The protractor shall be made of stainless or corrosion resistant steel, heat treated to produce a protractor capable of long life under service conditions.

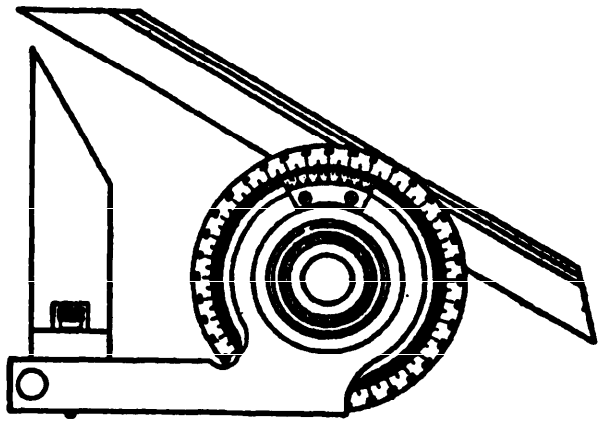


Figure 1. Protractor, universal bevel, acute-angle attachment, revolving dial.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: General Services Administration, Federal Supply Service, Tools and Appliances Commodity Center, Kansas City, MO 64131.

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3.1.1 Vernier scale. The vernier scale shall be made to fit neatly on the beveled edge of the frame and shall be held rigidly in position. The graduations shall extend to the outer surface edge of the scale and shall be made and fitted so as to minimize parallactic error in reading between the vernier scale and the main scale. There shall be a clearance of not more than 0.006 inch between the bottom surface of the vernier scale and the surface of the main scale directly beneath the vernier scales' outer lower edge. The vernier scale shall be graduated in 5-minute increments from 0 minutes up to 60 minutes both right and left from the zero index. The graduations shall be cut clearly and straight to a width of not less than 0.003 inch nor more than 0.006 inch, and to a depth of not less than 0.001 inch nor more than 0.003 inch. The 0 minute and each succeeding 15-minute graduation shall be marked with the corresponding number and shall be longer than the intermediate graduations. The intermediate graduations shall be readable to 5-minute increments. The vernier scale shall have an accuracy within ± 1 minute between any two graduations. In addition, each graduation marking shall be within 10 minutes of its true reading.

3.1.2 Acute-angle attachment. The acute-angle attachment shall fit readily on the base, and shall be capable of being clamped at any desired point within the limit provided in the base. When clamped, the longitudinal axis shall be perpendicular within ± 2 minutes to the base and in the same plane as the blade.

3.1.3 Frame. The frame of the protractor shall be of one-piece construction and so constructed that it will lie flat upon the work. The protractor shall consist essentially of a circular member upon which is mounted the vernier scale with a base whose top and bottom surface are at right angles to the diameter of the circular section passing through zero graduation on the vernier scale. At a zero setting, measuring edges of blades and base shall be parallel within ± 2-1/2 minutes. The measuring edges shall be flat within 0.0005 inch over the length of the edges. The frame shall have a not less than 81 HRC when tested in accordance with ASTM E18.

3.1.4 Dial. The dial shall be not less than 2-1/2 inches nor more than 3 inches in diameter and shall have a scale which is accurately graduated in degrees on the outer periphery of the front surface, and shall be rotatable in a complete circle upon a center stud. The scale shall be divided into four quadrants, each having 90 graduations. The zero and each 10-degree graduation in either direction shall be marked with the corresponding numeral. The 5-degree graduations shall be longer than the intermediate graduations. A tolerance of ± 2-1/2 minutes non-cumulative between any two graduations shall be allowed. The width and depth of graduation lines shall be as specified for the vernier scale (see 3.1.1).

3.1.5 Thumb nuts. Thumb nuts or other devices shall be furnished for locking the rotating dial to the frame and locking the blade at any point.

3.1.6 Setting. A positive means shall be furnished for securing fine adjustments and smooth movement of the dial in either direction.

3.1.7 Adjustable blades. Adjustable blades shall be capable of being moved smoothly back and forth through their entire length and shall be interchangeable. The blades shall have a longitudinal groove parallel with the edges to facilitate the clamping to the revolving dial and shall have a hardness value of not less than 92 HRC when tested in accordance with ASTM E18. The ends of the blades shall be machined to angles with one blade edge of 45 degrees on one end and 60 degrees on the other end. Blades of 12 inches, or 6-1/2 inches, ± 1/2 inch shall be furnished as specified (see 7.2). The blades shall be flat and parallel within 0.0005 inch.

3.1.8 Instruction leaflet. Each protractor shall be accompanied with a clear and concise operational instruction leaflet.

3.2 Identification marking. The tool shall be marked with the manufacturer's name or identifying symbol and the state or country of manufacture, unless otherwise specified. All identification markings shall be engraved, etched, molded, or indented directly on the item's surface in such a manner that it remains clearly legible throughout the life of the item.

3.3 Workmanship. Details of workmanship shall be in accordance with the best commercial practice. Paints, coatings, platings, and finishes shall be smooth, adherent, continuous, and not stained or discolored. External surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections. The item shall be free from



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manufacturing workmanship defects (e.g., sharp or rough external edges, corners, or surfaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety, or appearance.

4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified (using conversion tables contained in the latest revision of Federal Standard 376) and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

5. QUALITY ASSURANCE PROVISIONS.

5.1 Product conformance. The product provided to the Government shall conform to the contractually specified requirements. For product characteristics and requirements which are not contractually specified, the provided product shall conform to the producer's own drawings, standards, specifications, and quality assurance practices for the highest quality product commercially offered by the producer. If the product is not commercially offered by the producer, the best commercial practices shall apply. The Government reserves the right to require proof of such conformances prior to first delivery and thereafter as provided for under the provisions of the contract.

5.2 Responsibility for inspection. Unless otherwise specified, the contractor is responsible for the performance of all inspection requirements and may use any commercial facilities (including the contractor's own facilities) suitable for performance of the inspection requirements, unless disapproved by the Government. The Government reserves the right to perform any inspections deemed necessary to assure the item conforms to the specified requirements.

5.3 Bid sample(s). When a bid sample requirement is specified in the solicitation, the bid sample(s) shall be inspected for all salient characteristics by the Government. A failure of any bid sample to meet the salient characteristics shall be cause for rejection.

5.4 First article sample(s). When a first article requirement is specified in the contract, the first article sample(s) shall be inspected for all salient characteristics. A failure of any first article sample to meet the salient characteristics shall be cause for rejection.

5.5 Test procedures. Tests for accuracy shall be made with suitable equipment of proven accuracy. All tests shall be performed at 68 degrees F, ± 10 degrees F. All tests shall be performed at a constant temperature.

5.5.1 Flatness. The flatness shall be determined with a test indicator in conjunction with a surface gauge and surface plate.

5.5.2 Squareness. The squareness shall be determined by means of a precision square and gauge blocks.

5.5.3 Angular intervals. Angular intervals shall be determined by means of either a circular measuring machine, a sine bar, or angular blocks to ascertain the total angular error.

6. PACKAGING.

6.1 Packaging. Requirements of preservation, packing, packaging, and marking of packages shall be as specified in the contract or order.



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7. NOTES.

(This section contains information of a general or explanatory nature that is not mandatory.)

7.1 Addresses for obtaining copies of referenced documents:

Federal Specifications and Standards: Federal Supply Service Bureau, Specification Section, Suite 8100, 470 L'Enfant Plaza, SW, Washington, DC 20407.

Federal Acquisition Regulations (FAR): Government Printing Office, Superintendent of Documents, Washington, DC 20402-9371.

ANSI/ASQC Z1.4: American Society for Quality Control, P.O. Box 3005, 611 East Wisconsin Avenue, Milwaukee, WI 43201-4606.

7.2 Ordering data. Purchasers should select the preferred options permitted herein and should include the following information in procurement documents:

(a) Title, number, and date of this Commercial Item Description.
(b) When applicable, the appropriate NSN(s).
(c) If bid samples are required, the number of samples required, and the subjective inspection requirements.
(d) If first article samples are required and the number of samples required.
(e) If lot sampling inspection is required and the applicable lot sampling requirements. For example:

Sampling for inspection. Sampling for inspection (both examinations and testing) shall be in accordance with ANSI/ASQC Z1.4.

Inspection. Each sample tool shall be examined and tested for all salient characteristics. The Inspection Level shall be S-2 with an Acceptable Quality Level of 4.0, expressed in terms of defects per hundred units.

(f) Size of blades required (see 3.1.7).
(h) What preservation, packing, packaging, and marking of packages are required.

7.3 National Stock Numbers (NSNs). The following list of NSNs corresponds to the protractors specified in this CID. This list may not be indicative of all possible NSNs associated with this CID.

NSN

5210-00-273-3422

MILITARY INTERESTS:

NONE: DoD has no registered interest in revisions or notices to this Commercial Item Description until further notice.

PREPARING ACTIVITY:

GSA - FSS



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